

Software Requirements Specification

Project Name: ARCC (AI Resume and Career Coach)

Course: ENSF 400

Section: 01

Group #: 13

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1. Introduction

1.1 Scope

The AI Resume & Career Coach is a web-based application designed to help job seekers optimize their applications for specific job roles. Many candidates struggle to tailor resumes, identify missing skills, and practice interviews effectively. This system addresses these problems through automated resume matching, AI-driven gap analysis, and an interactive voice-based interview simulator that provides real-time feedback. The goal is to improve candidate preparedness, confidence, and application success rates.

1.2 Intended Audience

This document is intended for multiple stakeholders involved in the project:

- **Clients/End Users**—Individuals seeking employment who will use the platform to improve their resumes and better align with job roles.
- **Project Managers**—Responsible for overseeing development progress, scope, and deliverables.
- **Developers**—Engineers implementing the system architecture, backend logic, and AI components.
- **Quality Assurance (QA) Staff**—Personnel responsible for testing the system to ensure accuracy, reliability, and usability.
- **Stakeholders/Instructors (if academic)**—Individuals evaluating the system's design, functionality, and alignment with project objectives.

1.3 Definitions, Acronyms, Abbreviations

LLM: Large Language Model

STT: Speech-to-Text

UI: User Interface

API: Application Programming Interface

ATS: Applicant Tracking System

NLP: Natural Language Processing

2. Overall Description

2.1 Product Perspective

The system will function as a standalone web application. It will integrate external AI services for language processing and speech recognition. The application consists of a frontend user interface, a backend server, a database, and AI/LLM services. The system will integrate an external AI service using APIs, most likely OpenAI or Gemini.

2.2 User Classes and Characteristics

Job Seeker (Primary User):

Individuals using the system to improve resumes and applications. They may have limited technical knowledge, be goal-oriented, expect clear feedback, and want quick results.

Developer (Maintenance Role):

Responsible for updating features, fixing bugs, and improving AI integration. High technical expertise; interacts with backend, APIs, and database systems.

Admin (System Administrator):

Manages system operation, monitors performance, and handles maintenance. Technical background, understands system configuration and data management.

2.3 Operating Environment

The system will operate in the following environment:

- **Platform:** Web-based application
- **Access Devices:** Laptops and desktops
- **Supported Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge, Safari
- **Backend Environment:** Cloud-hosted server (e.g., AWS, GCP, or similar)
- **Programming Technologies:**
 - **Frontend:** React: HTML, CSS, JavaScript
 - **Backend:** Python Flask
 - **Database:** MySQL

- **External Services:** AI/LLM, most likely ChatGPT, API provider for gap analysis and text processing

2.4 Design and Implementation Constraints

The project is subject to the following constraints:

- **Technology Constraints:** Must use web technologies and AI APIs; performance depends on third-party LLM services.
- **Budget Constraints:** Limited access to paid AI API calls may restrict usage volume or model size.
- **Time Constraints:** Features such as the interview simulator may be deprioritised due to the development timeline.
- **Data Privacy:** User resumes contain personal data and require correct security logic.
- **Internet Dependency:** The system requires an active internet connection for AI processing.

2.5 Assumptions and Dependencies

Assumptions:

- Users have resumes in digital format (PDF or DOCX).
- Users understand basic job description terminology.
- AI models will produce reasonably accurate skill-gap analysis.

Dependencies:

- Availability and reliability of external LLM API services
- Stable hosting infrastructure
- Browser compatibility for frontend features
- Continued support for chosen programming frameworks

3. System Features

Feature ID: F-01

Feature Name: Resume Upload and Parsing

Description: This feature allows job seekers to upload their resumes in supported digital formats so the system can extract and analyze relevant content such as skills, experience, and education.

Priority: High

Functional Requirements:

- The system shall allow users to upload resumes in PDF and DOCX formats.
 - The system shall validate uploaded files for supported formats and size limits.
 - The system shall extract textual content from uploaded resumes for further analysis.
 - The system shall store uploaded resumes securely for the duration of the user session.
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Feature ID: F-02

Feature Name: Job Description Input

Description: This feature enables users to input or paste a job description that they are applying for, which will be used as a reference for resume matching and gap analysis.

Priority: High

Functional Requirements:

- The system shall allow users to enter a job description using a text input field.
- The system shall validate that the job description is not empty before analysis.

- The system shall store the job description temporarily for comparison with the resume.
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Feature ID: F-03

Feature Name: Resume-to-Job Matching Analysis

Description: This feature compares the user's resume against the provided job description using AI-based text analysis to evaluate relevance and alignment.

Priority: High

Functional Requirements:

- The system shall analyze the resume content against the job description.
 - The system shall identify overlapping skills, keywords, and qualifications.
 - The system shall generate a matching score indicating resume alignment with the job role.
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Feature ID: F-04

Feature Name: Skill Gap Identification

Description: This feature identifies missing or weak skills in the user's resume relative to the job description and provides insights to improve job readiness.

Priority: Medium

Functional Requirements:

- The system shall identify skills present in the job description but missing from the resume.
- The system shall categorize identified gaps into technical and non-technical skills.
- The system shall present the skill gaps in a clear and readable format.

Feature ID: F-05

Feature Name: Resume Improvement Suggestions

Description: This feature provides AI-generated suggestions to help users improve their resumes by addressing identified gaps and enhancing clarity and relevance.

Priority: Medium

Functional Requirements:

- The system shall generate improvement suggestions based on the skill gap analysis.
 - The system shall provide actionable feedback, such as skill additions or wording improvements.
 - The system shall ensure suggestions are aligned with the selected job description.
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Feature ID: F-06

Feature Name: User Account Management

Description: This feature allows users to create and manage accounts to save resumes, job descriptions, and analysis results.

Priority: Low

Functional Requirements:

- The system shall allow users to register using an email address and a password.
 - The system shall authenticate users during login.
 - The system shall allow users to view previously uploaded resumes and analyses.
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Feature ID: F-07

Feature Name: Interview Practice Simulator

Description: This feature provides users with an AI-driven interview practice experience using text or voice-based interaction and feedback.

Priority: Low

Functional Requirements:

- The system shall generate interview questions based on the job description.
 - The system shall allow users to respond using text or voice input.
 - The system shall provide feedback on responses to help users improve interview performance.
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4. External Interface Requirements

4.1 User Interfaces

The AI Resume & Career Coach (ARCC) system will be accessed through a web-based user interface designed for ease of use by job seekers with minimal technical knowledge. All pages will include consistent navigation elements such as a header menu, back buttons, and logout options for authenticated users.

The main UI includes:

- **Home Page**

The page introduces the system's purpose and provides basic instructions on how to navigate to the login page.

- **Login/Registration Page**

The page allows users to create an account or authenticate using an email address and password.

- **Resume Upload Page**

This page allows users to upload resumes in PDF or DOCX format. It displays validation messages for unsupported files and confirms successful uploads.

- **Job Description Input Page**

Provides a text input area where users can paste or enter a job description for analysis.

- **Analysis Results Page**

The results page displays the resume-to-job matching score, identified skill gaps, and presents AI-generated resume improvement suggestions.

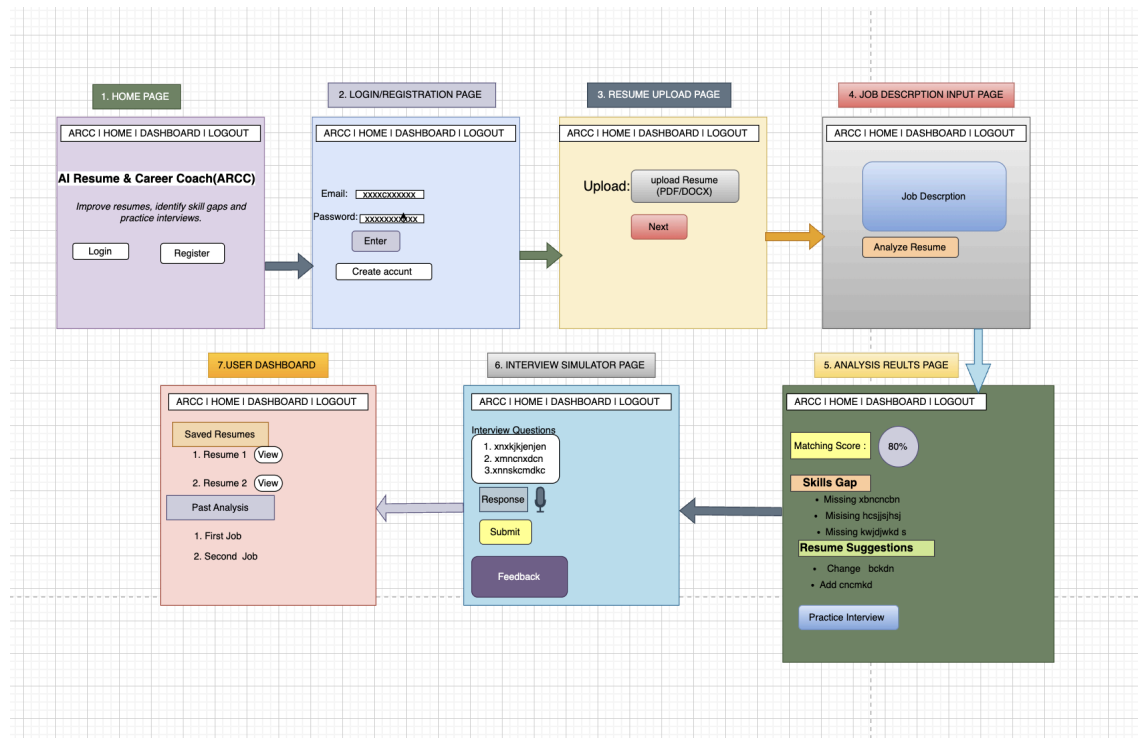
- **Interview Simulator Page**

This interface presents AI-generated interview questions and allows users to respond using text or voice input, with feedback provided after responses.

- **User Dashboard**

Displays previously uploaded resumes, job descriptions, and analysis results for authenticated users.

Figure 1: ARCC User Interface Sketch



This sketch illustrates the main screens, navigation flow, and key UI components of the ARCC System.

4.2 Software Interfaces

The ARCC system will interact with the following software components:

- **Database Interface**

The system shall store user accounts, resumes, job descriptions, and analysis results in a relational database (MySQL).

- **AI/LLM Service Interface**

The system shall integrate with an external AI service through APIs to perform resume analysis, skill gap identification, and content generation.

- **Speech-to-Text Interface**

The interview simulator shall integrate with a speech-to-text service to process voice-based user responses.

- **Backend API Interface**

The frontend client shall communicate with the backend server using RESTful APIs to submit data and retrieve results.

4.3 Communication Interfaces

- The ARCC system utilizes standard web-based communication protocols to support interaction between internal and external system components.
- Secure HTTPS connections are used for all data exchanges between the frontend and backend to ensure confidentiality and data integrity.
- The backend server communicates with external artificial intelligence services through RESTful APIs.
- JSON is used as the standard data interchange format for all requests and responses.
- Near real-time communication is required for the interview simulator feature to enable interactive question-and-response sessions.
- An active internet connection is required for all AI-dependent processing and external service interactions.

5. Non-Functional Requirements

Performance

- Response Latency/Time:
 - Analysis of one's resume: Dissect the resume and gap
 - Voice Simulator: The interview simulator must have a latency of less than 200 msec for processing audio to prevent lags during a conversation and encourage a less robotic dialogue flow
- Optimization for low-end devices: All AI processing must be done server-side, and the app should be lightweight, especially devices with limited RAM like Chromebooks or budget smartphones or old laptops
- Connectivity: A strong internet connection is required, but if the connection drops during an interview simulation, the user's system should locally cache the current state of the interview and attempt to reconnect automatically

Security

- All user data must be encrypted using TLS 1.3 and using AES-256 standards, mostly for resume files and recordings of interviews
- This web application must be protected against the OWASP Top 10 vulnerabilities with significant safeguards against SQL injections and cross-site scripting.
- Anonymized data should be processed by the AI model and voice recordings for the interview simulation(s) should be stored for the duration of the session and then deleted after it's over.

Usability

- Accessibility is a must; the platform must align with WCAG 2.1 Level AA standards, which include screen readers compatible with resume results and a high-contrast setting for the visually impaired.
- Responsiveness on mobile devices and interview features must also be functional.
- Feedback loops are essential for the user to understand decisions made by the AI in order to prevent bias and "black box" confusion

Reliability

- The availability of the app is important since the system will target 99.9% uptime during business hours from 8 AM to 8 PM across major time zones to support active job searches.

- The system should execute “all-or-nothing” database operations (known as atomic transactions) so if a resume has failed to upload halfway, there should be no partial/corrupted records in the database
- In the events of server crashes (during an interview simulation), the system should save the user’s progress up until the last saved question, allowing them to resume

Scalability

- Horizontal scaling in the backend
- The database must handle at least 10,000 new resume records per month without performance degradation
- The system must support at least 1,000 concurrent users performing resume inspection simultaneously without exceeding the response time limit of 5 seconds

Other

- Preventing AI bias
- Compliance with GDPR and CCPA